OPD vs. DMD: the loss/gradient relationship

Pointwise velocity regression vs. distribution matching for flow-matching distillation, and why they coincide only in the infinite-capacity limit

Flow-Factory / XOPD notes

2026-07-27

Abstract

We make precise the relationship between the *pointwise* distillation loss used by DiffusionOPD / XOPD (velocity or clean-latent MSE, "`xopd_dk_space`") and the *distribution-matching* loss of DMD/DMD2. Writing both gradients in the shared flow-matching $\hat{x}_0$ parameterization, we show they carry the *same* error signal ($\hat{x}_0^\theta - \hat{x}_0^\psi$) but differ in (i) the Jacobian they backpropagate through — $\partial_\theta v_\theta$ (the velocity output at a fixed input) for OPD vs. $\partial_\theta x$ (the generated sample, through the sampler) for DMD — and (ii) whether the "fake" term is an *independent, online* score of the student distribution. The two objectives share an optimum under infinite capacity, but under a capacity gap the L^2-optimal field differs from the KL-optimal distribution. A corollary: DMD with the fake pinned to the teacher, or to the student's own one-step map, collapses to a (reweighted) MSE; genuine distribution matching requires the third, online fake-score network. This is exactly the piece the self-normalized `x0_norm` space structurally lacks.

1 Setup and notation

Rectified-flow / flow matching on a shared latent space. For a clean latent x_0, noise $\varepsilon \sim \mathcal{N}(0, I)$, and schedule $\sigma = \sigma(t) = t/T \in [0, 1]$,

$$x_t = (1-\sigma)\, x_0 + \sigma\, \varepsilon, \qquad v = \varepsilon - x_0, \qquad \hat{x}_0(x_t) = x_t - \sigma\, v(x_t), \tag{1}$$

so the velocity v and the one-step clean estimate $\hat{x}_0$ are interchangeable. The (frozen) teacher velocity field is v_ψ, the student is v_θ; write $\hat{x}_0^\theta = x_t - \sigma v_\theta$, $\hat{x}_0^\psi = x_t - \sigma v_\psi$, hence

$$\hat{x}_0^\theta - \hat{x}_0^\psi = -\sigma\, (v_\theta - v_\psi). \tag{2}$$

The student induces a sample distribution p_θ (its ODE/consistency rollout); the teacher/target is p_{real}.

2 OPD: pointwise field regression

DiffusionOPD / XOPD's per-step term d_k is a plain MSE in one of four spaces (`xopd_dk_space`$\in \{v,\, xt,\, x0,\, x0\_norm\}$). Using (**??**), at a *given* x_t,

$$L^{(v)} = \mathbb{E}\, \|v_\theta - v_\psi\|^2, \qquad \nabla_\theta L^{(v)} = 2\, \mathbb{E}\big[(v_\theta - v_\psi)\, \partial_\theta v_\theta\big], \tag{3}$$

$$L^{(x_0)} = \mathbb{E}\, \left\|\hat{x}_0^\theta - \hat{x}_0^\psi\right\|^2 = \mathbb{E}\, \sigma^2\, \|v_\theta - v_\psi\|^2, \qquad \nabla_\theta L^{(x_0)} = 2\, \mathbb{E}\big[\sigma^2 (v_\theta - v_\psi)\, \partial_\theta v_\theta\big], \tag{4}$$

$$L^{(xt)} = \mathbb{E}\, \|\mu_\theta - \mu_\psi\|^2 \propto \mathbb{E}\, \Delta t^2\, \|v_\theta - v_\psi\|^2, \qquad \nabla_\theta L^{(xt)} = 2\, \mathbb{E}\big[\Delta t^2 (v_\theta - v_\psi)\, \partial_\theta v_\theta\big], \tag{5}$$

where $\mu = x_t + v\,\Delta t$ is the Euler transition mean. Thus the three "geometric" spaces are one gradient reweighted by $a(t) \in \{1, \Delta t^2, \sigma^2\}$ (the identity $\mathrm{MSE}(v) : \mathrm{MSE}(xt) : \mathrm{MSE}(x_0) = 1 : \Delta t^2 : \sigma^2$). The self-normalized space divides the x_0 loss by a detached per-sample scale,

$$L^{(x0\_norm)} = \mathbb{E}\left[\frac{\left\|\hat{x}_0^\theta - \hat{x}_0^\psi\right\|^2}{\mathrm{sg}\left(|\hat{x}_0^\theta - \hat{x}_0^\psi|\right) + \epsilon}\right], \qquad \nabla_\theta L^{(x0\_norm)} = \mathbb{E}\left[\frac{2\sigma\,(v_\theta - v_\psi)\,\partial_\theta v_\theta}{\mathrm{sg}(\mathrm{scale})}\right], \tag{6}$$

which equalizes each step's gradient magnitude (DiffusionNFT/DMD-style adaptive reweighting) but does *not* change the direction. **Two structural facts hold for every OPD variant:**

1. the *target* is the teacher *field* v_ψ (equivalently $\hat{x}_0^\psi$) evaluated at the *same* input x_t; the error signal is $(v_\theta - v_\psi) \propto (\hat{x}_0^\theta - \hat{x}_0^\psi)$;
2. the gradient backpropagates through $\partial_\theta v_\theta$ — the velocity *output* at a *fixed* input x_t. It is behaviour-cloning of a vector field; the sampler is never differentiated.

3 DMD: distribution matching

DMD trains a generator $x = G_\theta(z, c)$ (few-step) to minimize $\mathrm{KL}(p_\theta \,\|\, p_{\mathrm{real}})$ via the variational-score-distillation gradient. With a Gaussian perturbation at level t, the score and the one-step clean estimate are related by $\nabla \log p_t(x_t) = -(x_t - \hat{x}_0^p)/\sigma^2$, so the score *difference* between the target and the student's own distribution is

$$s_{\mathrm{real}}(x_t) - s_{\mathrm{fake}}(x_t) \;\propto\; \frac{\hat{x}_0^{\mathrm{fake}}(x_t) - \hat{x}_0^{\mathrm{real}}(x_t)}{\sigma^2}, \tag{7}$$

where $\hat{x}_0^{\mathrm{real}}$ comes from the frozen teacher and $\hat{x}_0^{\mathrm{fake}}$ from an *online* score network trained (by denoising) to track the current p_θ. The DMD gradient is

$$\nabla_\theta L_{\mathrm{DMD}} = \mathbb{E}_{t,\, x = G_\theta(z)}\Big[w(t)\,\big(s_{\mathrm{fake}}(x_t) - s_{\mathrm{real}}(x_t)\big)\,\partial_\theta x\Big]. \tag{8}$$

The DMD2 implementation realizes (**??**) in $\hat{x}_0$ space with a stop-gradient identity. With $p_{\mathrm{real}} = x - \hat{x}_0^{\mathrm{real}}$, $p_{\mathrm{fake}} = x - \hat{x}_0^{\mathrm{fake}}$, and the self-normalization norm $= \overline{|p_{\mathrm{real}}|}$,

$$g = \frac{p_{\mathrm{real}} - p_{\mathrm{fake}}}{\mathrm{norm}} = \frac{\hat{x}_0^{\mathrm{fake}} - \hat{x}_0^{\mathrm{real}}}{\mathrm{norm}}, \qquad L_{\mathrm{DMD}} = \tfrac{1}{2}\,\|x - \mathrm{sg}(x - g)\|^2 \;\Rightarrow\; \nabla_\theta L_{\mathrm{DMD}} = g\,\partial_\theta x. \tag{9}$$

Two structural facts: (i) the direction is the *difference of two scores* $\hat{x}_0^{\mathrm{fake}} - \hat{x}_0^{\mathrm{real}}$, moving x *away from* its own current density and *toward* the target; (ii) the gradient backpropagates through $\partial_\theta x$ — the generated *sample* position, i.e. through the sampler.

4 The core relationship

Both gradients are "(an $\hat{x}_0$ residual) $\times$ (a Jacobian)". Table **??** lines them up.

4.1 Differentiate w.r.t. v or w.r.t. x

This is the crux the two objectives disagree on. OPD moves θ so the velocity *output* $v_\theta(x_t)$ matches $v_\psi(x_t)$ at inputs x_t drawn from the rollout, holding x_t fixed in the backward pass. DMD moves θ so the *sample* $x = G_\theta(z)$ shifts along $\hat{x}_0^{\mathrm{real}} - \hat{x}_0^{\mathrm{fake}}$, differentiating through the sampler ($\partial_\theta x$). Field-matching vs. sample-moving.

	OPD (MSE(x_0)/*x0_norm*)	**DMD**
error signal	$\hat{x}_0^\theta - \hat{x}_0^\psi$ (student−teacher, one point)	$\hat{x}_0^{\text{fake}} - \hat{x}_0^{\text{real}}$ (fake−real scores)
Jacobian	$\partial_\theta \hat{x}_0^\theta = -\sigma\, \partial_\theta v_\theta$ (velocity output)	$\partial_\theta x$ (generated sample, thru sampler)
"fake" role	— (none)	*independent online* score of p_θ
metric minimized	L^2 on the velocity *field*	KL on the induced *distribution*
optimum	conditional-mean field	mode-seeking / sharper samples
stability	stable, single fixed target	adversarial-flavored, two-timescale

Table 1: OPD and DMD carry an $\hat{x}_0$-residual times a Jacobian; they differ in the residual's second term, the Jacobian, and whether a third online network is present.

4.2 Reduction 1: fake = teacher ⇒ no signal

If the online fake is pinned to the teacher, $\hat{x}_0^{\text{fake}} = \hat{x}_0^{\text{real}}$ and (**??**) gives $g = 0$: DMD is *degenerate*. Note this is *not* what *x0_norm* does: *x0_norm* keeps the *OPD* residual $\hat{x}_0^\theta - \hat{x}_0^\psi$ (student−teacher) and only borrows DMD2's per-sample self-normalization. So

$$\boxed{\begin{array}{c} x0\_norm = \text{OPD } x_0 \text{ residual } + \text{ DMD2 self-normalization;} \\ \text{it is } \textit{not} \text{ distribution matching (no online fake, differentiates } v, \text{ not } x). \end{array}}$$

4.3 Reduction 2: fake = the student's own one-step map ⇒ reweighted MSE

Suppose one "cheaply" sets $\hat{x}_0^{\text{fake}} = \hat{x}_0^\theta$ (the student's own one-step clean prediction) and $\hat{x}_0^{\text{real}} = \hat{x}_0^\psi$. Then the DMD direction becomes the OPD residual,

$$g = \frac{\hat{x}_0^{\text{fake}} - \hat{x}_0^{\text{real}}}{\text{norm}} = \frac{\hat{x}_0^\theta - \hat{x}_0^\psi}{\text{norm}}, \tag{10}$$

and (**??**) gives $\nabla_\theta L = g\, \partial_\theta x$, versus OPD's $\nabla_\theta L^{(x_0)} = (\hat{x}_0^\theta - \hat{x}_0^\psi)\, \partial_\theta \hat{x}_0^\theta = -(\hat{x}_0^\theta - \hat{x}_0^\psi)\, \sigma\, \partial_\theta v_\theta$. *Same error signal, different Jacobian* ($\partial_\theta x$ vs. $-\sigma\, \partial_\theta v_\theta$).

Why "biased" (not a pejorative on MSE). True DMD needs $s_{\text{fake}} = \nabla \log p_\theta$, the score of the *distribution of samples* produced by the generator G_θ (the pushforward of the noise prior under the sampler). Setting $\hat{x}_0^{\text{fake}} = \hat{x}_0^\theta(x_t)$ substitutes a different object: the student's *pointwise field* evaluated at x_t. These coincide only when v_θ is consistent with being the MMSE denoiser / score of its own induced p_θ (infinite capacity + consistency). In general — and especially for a capacity-limited or few-step student being distilled toward v_ψ — v_θ approximates the *teacher field*, not $\nabla \log p_\theta$. So $\hat{x}_0^\theta$ is a **biased estimator of the fake score**: the KL identity fails even though the residual looks like an MSE. What remains is precisely a (self-)reweighted OPD residual, only routed through $\partial_\theta x$ instead of $\partial_\theta v_\theta$.

Proposition (No free lunch)**.** *Replacing $\partial_\theta v_\theta$ by $\partial_\theta x$ (routing the same MSE residual through the sampler), or pinning the fake to teacher/student, does* not *produce a distribution loss. Equation* (**??**) *equals the* KL *gradient only when $s_{\text{fake}} = \nabla \log p_\theta$ is an* independent, freshly trained *estimate of the student's own (moving) score. With fake=teacher the signal vanishes; with fake=student ($\hat{x}_0^{\text{fake}} \leftarrow \hat{x}_0^\theta$) the fake score is biased for $\nabla \log p_\theta$, leaving a reweighted MSE routed through the rollout. Hence genuine DMD is irreducibly a* three-network *system (real, online-fake, student); it cannot be obtained from OPD by an algebraic substitution.*

4.4 Why they agree only in the infinite-capacity limit

If the student can represent the teacher field exactly, the optimum $v_\theta^\star = v_\psi$ makes *both* gradients vanish (OPD residual = 0; then $p_\theta = p_{\text{real}}$ so the DMD score difference = 0): the objectives share this optimum, and pointwise matching a *deterministic* teacher field is the principled thing to do (it is not the "mode-averaging" trap, because the per-input target is single-valued). Under a capacity gap the objectives diverge: OPD returns the L^2 projection onto the field (a conditional mean, whose *induced* distribution can be blurred), whereas DMD returns the KL-closest distribution (mode-seeking, sharper). The L^2 metric on the velocity field is simply not the KL metric on the transported distribution, so for a capacity-limited student (small model / few steps) DMD can produce better samples even when distilling the very same deterministic teacher.

5 Extensions (for completeness)

TDM (Trajectory Distribution Matching). Match p_θ to p_{real} not only at the endpoint but at several trajectory times $t \in \{0.75, 0.5, 0.25, 0\}$, so the gradient acts on the whole latent sequence rather than the final sample; this constrains the transport *dynamics*, not just the terminal density.

Transport-operator view. A flow is an operator $\Phi_{1\to 0}$. Merging teachers is a mixture of induced distributions $\Phi_S \approx \sum_i \pi_i \Phi_i$ (a distribution-level statement), and factorizing is a composition $\Phi_T \approx \Phi_K \circ \cdots \circ \Phi_1$; the linear velocity combination $v = \sum_i w_i v_i$ is only a first-order surrogate for these, which is why velocity-space MSE and image/trajectory-space distribution matching are genuinely different.

6 Practical implications for XOPD

- OPD's move `xt`$\to$`v/x0` fixed the per-step reweighting (removing the Δt^2 last-step dominance) and is stable and correct in the capacity limit; it is *not* "going backwards".
- `x0_norm` = OPD residual + DMD2 self-normalization: it equalizes per-step gradient magnitude but is still pointwise field regression — no online fake, differentiates v not x.
- To obtain a genuine distribution signal one must add the *online fake-score* network and differentiate through the sampler ($\partial_\theta x$), i.e. the `XDMDTrainer` design (`docs/xopd/dmd_cross_model_design.md`): real = 32B teacher, fake = a second LoRA adapter tracking p_θ, student = the few-step generator; direction $\hat{x}_0^{\text{fake}} - \hat{x}_0^{\text{real}}$, applied by the stop-gradient identity (**??**). That online fake is precisely the term `x0_norm` lacks.